



THE UNIVERSITY *of* EDINBURGH
Edinburgh Medical School

Tutorial 2

Data Types and Structures in Python and R

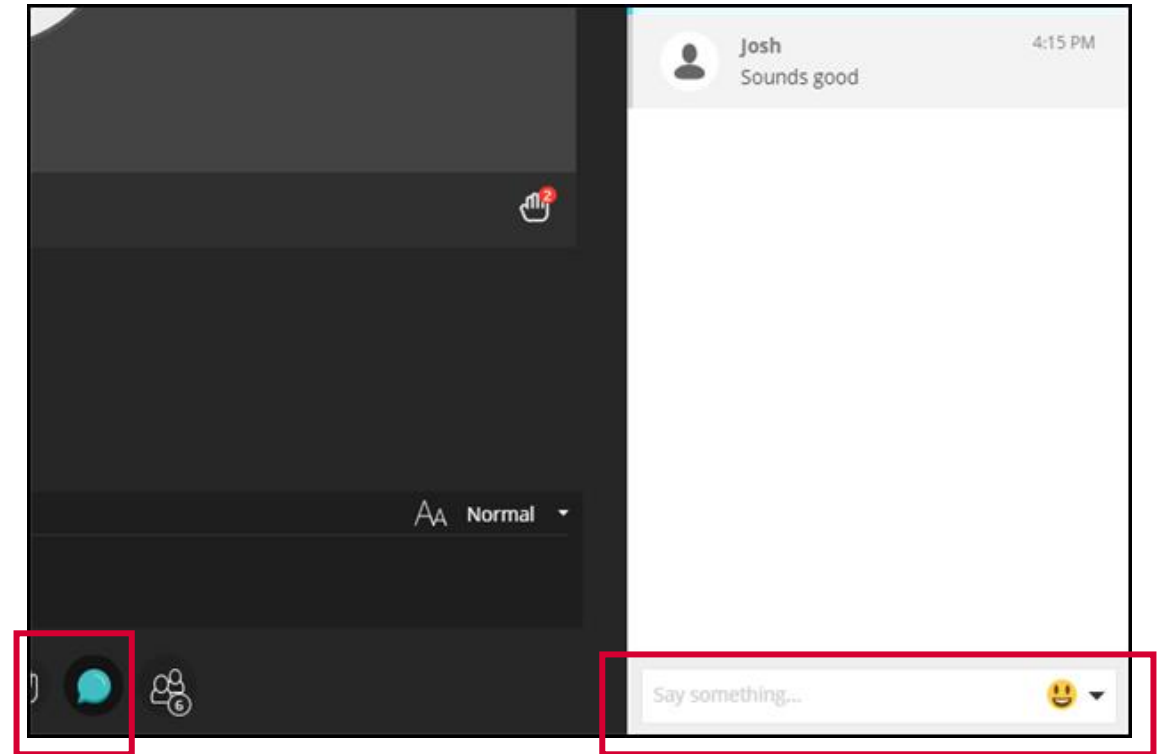
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Please type **yes** or **no** in the “Text chat area”

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- Type into the chat box and a moderator will try to assist you



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A black rectangular button with a white camera icon on the left and the text "Start Recording" in white on the right.

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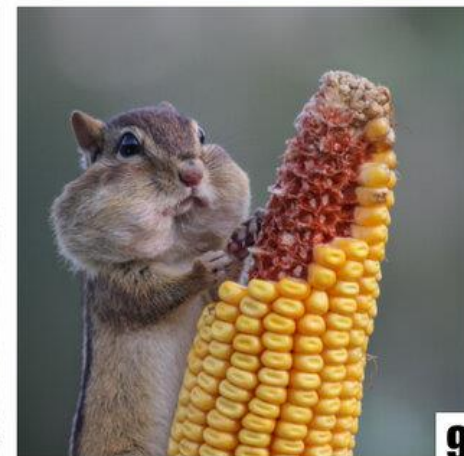


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Tutorial 2

Data Types and Structures in Python and R

How are you feeling today on this range of expressive animals?



Agenda

- Week 3 Drop in time
-  Error of the week
 -  Understanding and navigating folder structures in code
 -  Errors! Oh my ... how to read them
-  Data types overview
-  Data types activity
-  Any questions

Vote for Week 3 drop-in time!



Tuesday 2 June
4pm BST

OR



Wednesday 3 June
1pm BST

!?! Error of the week !?!

- 📁 Understanding and navigating folder structures in code
 - 📖 How to read error messages

How to find your current path

Python

- OS module to work with paths independent of operating systems
- `import os`
`os.getcwd()`
 - to get the current working directory

R

- `getwd()`
 - base R function to get the current working directory
- When using R Projects, your current working directory automatically points to the root folder where that .Rproj file is saved
- When using an RMD, it is where the RMD file is located

Navigating folder structures in code

File path = location of a file on a computer's file system structure

Absolute path = specified from the root directory (which is the first or topmost directory)

- AKA “full file paths”
- ~ commonly used to represent user's home directory
- e.g., `C:\Users\bblankin\Teaching\AY2025-26\Data Types and Structures in Python and R\Tutorial Slides` **or** `~\Teaching\AY2025-26\Data Types and Structures in Python and R\Tutorial Slides`

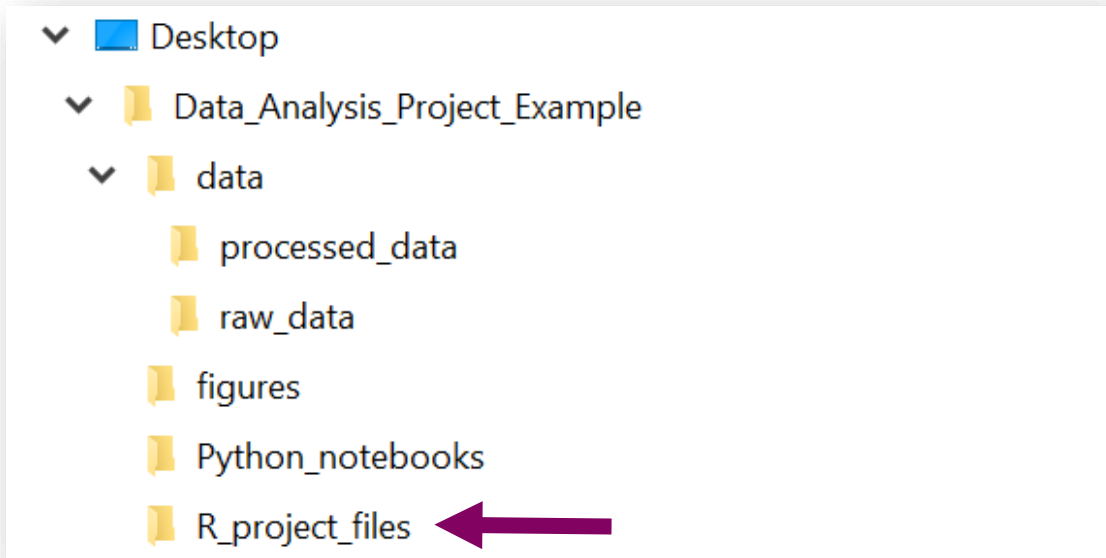
Relative path = path relative to the current directory

- Reproducible if you share your code with someone who has the same file and folder set up!
- Single dot (.) indicates current directory & double dot (..) represents parent directory
- e.g., `.\Data Types and Structures in Python and R\Tutorial Slides`

Path separators formatting note: Mac & Linux use / whereas Windows uses \

- URLs follow a standard format always using forward slash / regardless of operating system

Navigating folder structures in code



In Noteable, your home directory
(`\home\jovyan\` or `~\`) = Jupyter Notebook
File Browser (what you see when you open a
Standard Python 3 server)

The absolute path to a RMD file in “R_project_files” is
location:

```
~\Desktop\Data_Analysis_Project_Example\  
R_project_files\R_file.RMD
```

To navigate **up** the folder tree, use the “`..`” prefix.

To navigate 2 levels up, repeat the up prefix twice “`..\..\`”

For example, to go up to figures from R_project_files it would be:

```
..\figures\figure_1.png
```

To get to raw_data from within R_project_files:

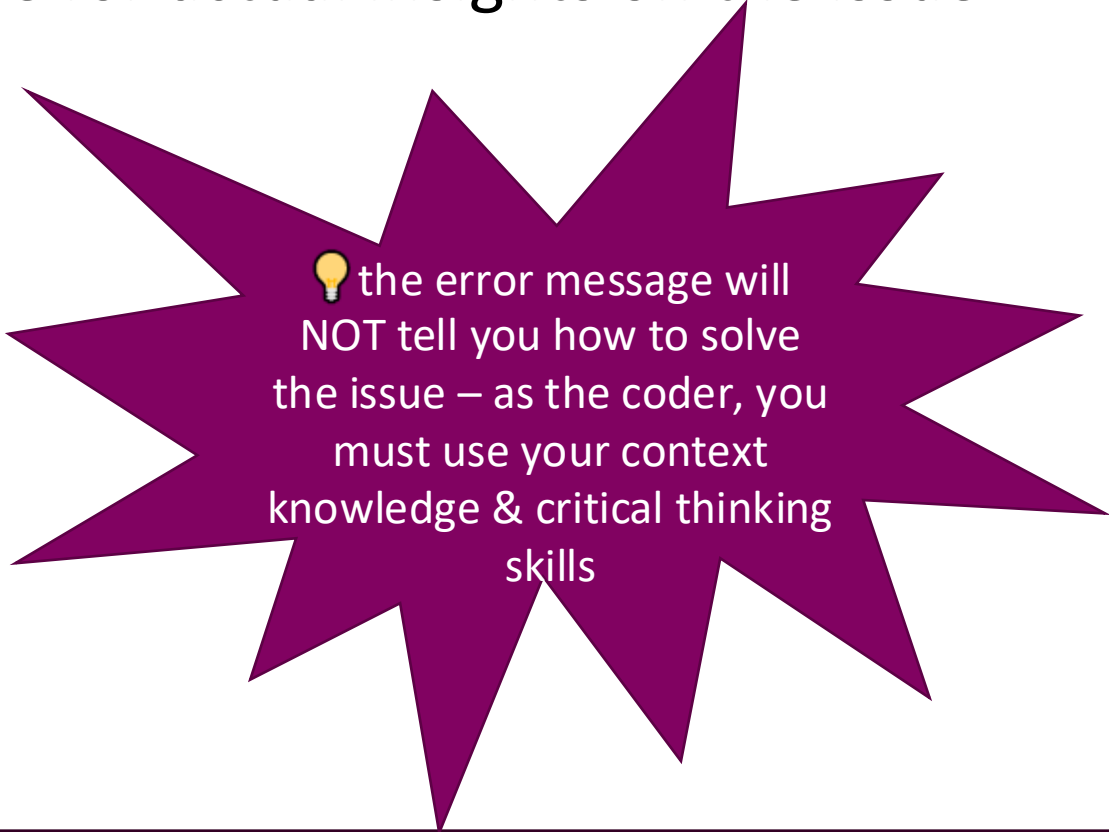
```
..\data\raw_data\raw_data.csv
```

How to read errors

- Step 1: Take a deep breath
- Step 2: Go to the bottom and read there for actual insights on the issue

 Read from bottom to top!

- Key components:
 - Error location (if applicable)
 - Error type (more likely in Python)
 - Error message



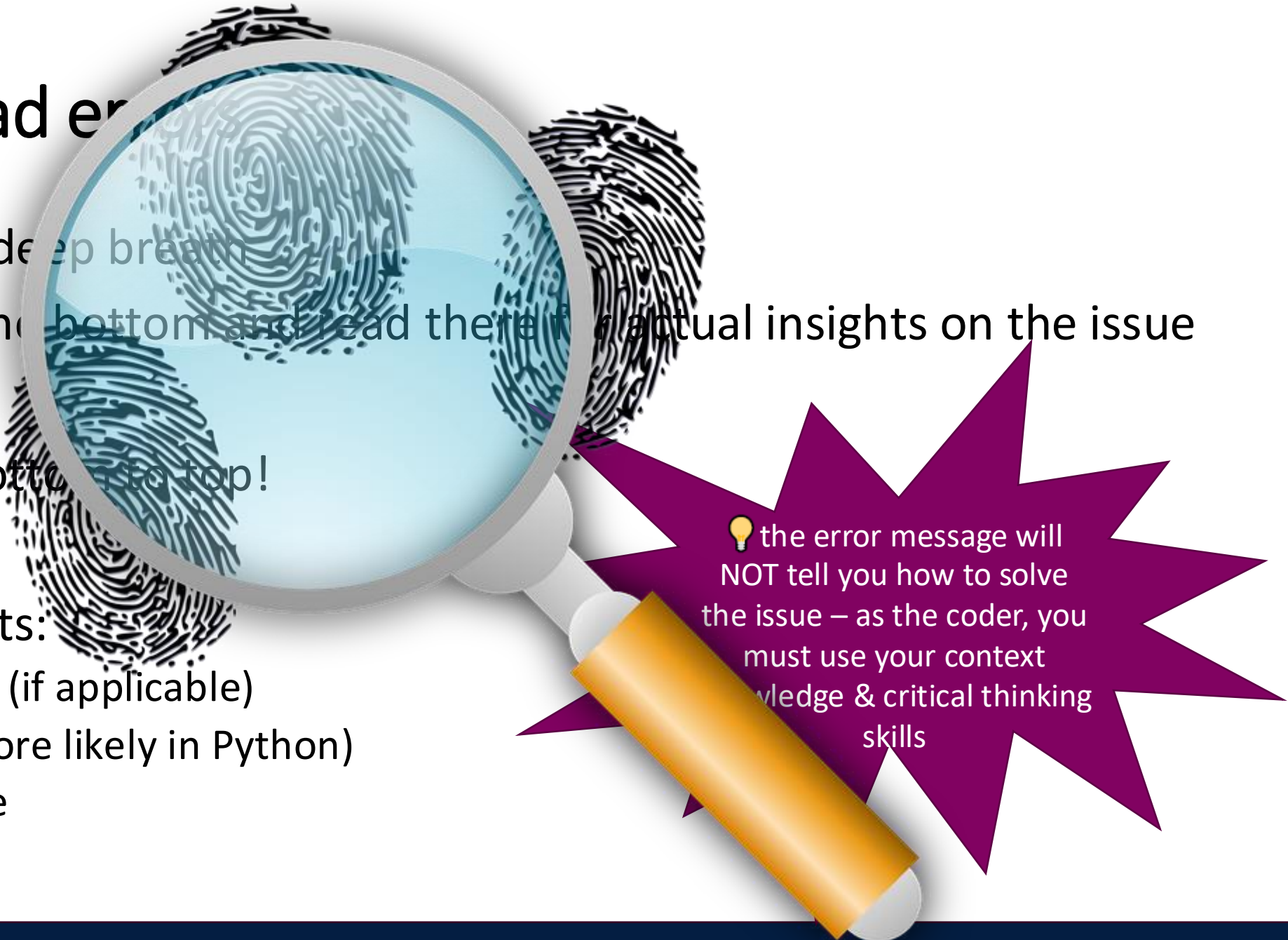
💡 the error message will NOT tell you how to solve the issue – as the coder, you must use your context knowledge & critical thinking skills

How to read error messages

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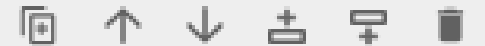


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Some examples – Python

[1]:

X



NameError

Traceback (most recent call last)

Cell In[1], line 1

----> 1 X

NameError: name 'X' is not defined

Error type

Error message

Error location

Some examples – Python

Error type

Error message

Error location

```
[5]: from IPython.display import Image
Image(filename = "figures/PythoLogo.png")

-----
FileNotFoundError                                Traceback (most recent call last)
Cell In[5], line 2
      1 from IPython.display import Image
----> 2 Image(filename = "figures/PythoLogo.png")

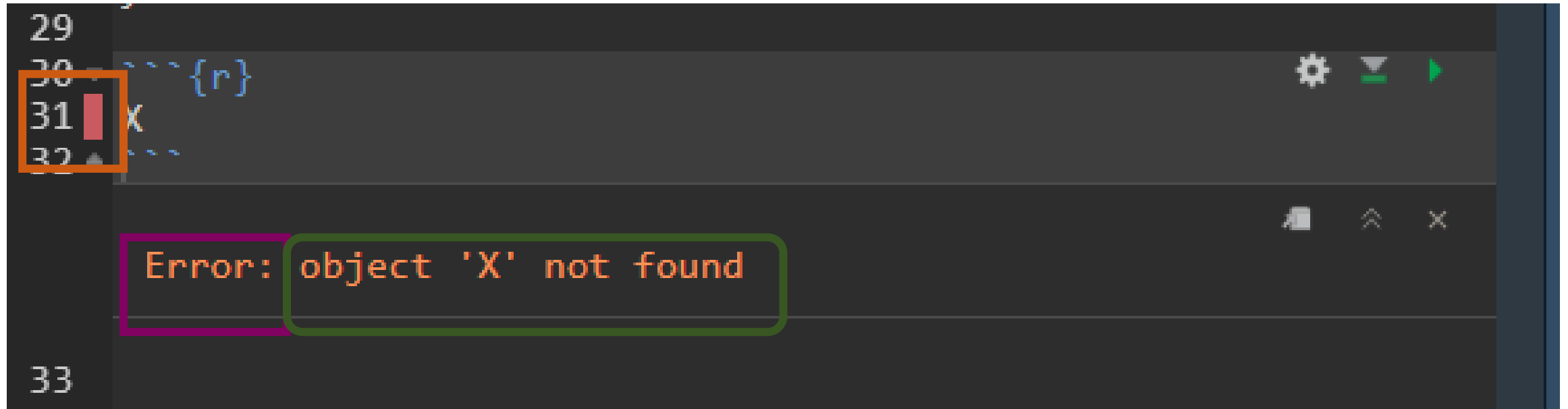
File /opt/conda/lib/python3.11/site-packages/IPython/core/display.py:970, in Image.__init__(self, data, url, filename, format, embed, width, height, retina, unconfined, metadata, alt)
    968 self.unconfined = unconfined
    969 self.alt = alt
--> 970 super(Image, self).__init__(data=data, url=url, filename=filename,
    971                             metadata=metadata)
    973 if self.width is None and self.metadata.get('width', {}):
    974     self.width = metadata['width']

File /opt/conda/lib/python3.11/site-packages/IPython/core/display.py:327, in DisplayObject.__init__(self, data, url, filename, metadata)
    324 elif self.metadata is None:
    325     self.metadata = {}
--> 327 self.reload()
    328 self._check_data()

File /opt/conda/lib/python3.11/site-packages/IPython/core/display.py:1005, in Image.reload(self)
   1003 """Reload the raw data from file or URL."""
   1004 if self.embed:
-> 1005     super(Image, self).reload()
   1006     if self.retina:
   1007         self._retina_shape()

File /opt/conda/lib/python3.11/site-packages/IPython/core/display.py:353, in DisplayObject.reload(self)
    351 if self.filename is not None:
    352     encoding = None if "b" in self._read_flags else "utf-8"
--> 353     with open(self.filename, self._read_flags, encoding=encoding) as f:
    354         self.data = f.read()
    355 elif self.url is not None:
    356     # Deferred import
FileNotFoundError: [Errno 2] No such file or directory: 'figures/PythoLogo.png'
```

Some examples – R (in RMD/console)



The screenshot shows an R console window with a dark background. Line numbers 29, 30, 31, and 32 are visible on the left. Line 31 contains a red vertical bar and the character 'X'. Below the code, an error message is displayed: "Error: object 'X' not found". The word "Error:" is highlighted with a purple box, and the rest of the message "object 'X' not found" is highlighted with a green box. The line number 33 is visible at the bottom left of the console.

Error type

Error message

Error location

Some examples – R (knitting)

Error type

Error message

Error location

The screenshot displays the RStudio interface with a file named 'Tutorial01-R.Rmd'. The left pane shows the source code of the R Markdown document, which includes a YAML header and an R code chunk. A warning message at the top of the source pane states: 'Package gapminder required but is not installed. Install Don't Sx'. The right pane shows the console output, which includes the error message: 'Error in `library()` : ! there is no package called 'gapminder''. Below the error message, a backtrace is shown, listing the functions called during the knitting process. At the bottom of the console, a message indicates: 'Quitting from lines 63-69 [load-packages] (Tutorial01-R.Rmd) Execution halted'. The bottom status bar shows 'Environment is empty'.

```
1 ---
2 title: "Tutorial 1: R and RMarkdown"
3 author: "YOUR NAME HERE"
4 date: "DATE HERE"
5 output:
6   html_document: default
7 ---
8
9 {r setup, include=FALSE}
10 knitr::opts_chunk$set(echo = TRUE)
11
12
13
14 R Markdown documents are comprised of 3
15 primary elements:
16 1. the YAML or metadata at the top
17
18 # Task 0
19
```

processing file: Tutorial01-R.Rmd

Error in `library()` :
! there is no package called 'gapminder'

Backtrace:

1. rmarkdown::render(...)
2. knitr::knit(knit_input, knit_output, envir = envir, quiet = quiet)
3. knitr::process_file(text, output)
6. knitr::process_group(group)
7. knitr::call_block(x)
- ...
14. base::withRestarts(...)
15. base (local) withRestartList(expr, restarts)
16. base (local) withOneRestart(withRestartList(expr, restarts[-nr]), restarts[[nr]])
17. base (local) doCall(restart\$handler, restartArgs)
19. evaluate (local) fun(base::quote(<pckgNtFE>))

Quitting from lines 63-69 [load-packages] (Tutorial01-R.Rmd)
Execution halted

Environment is empty

Some examples – R (knitting) ... what lines 63-69 look like when code is run in the RMD document

Error type

Error message

Error location

```
61
62 {r load-packages}
63 Sys.setenv(TZ="Europe/London") # set timezone (TZ argument) to whatever timezone you
   may be in, I am in the UK so am using the TZ identifier for London. The allows the to
   packages to work as intended
64 # for a list of TZ identifiers see here:
   https://en.wikipedia.org/wiki/List\_of\_tz\_database\_time\_zones
65
66 library(tidyverse)
67
68 library(gapminder) # for our data
69
```

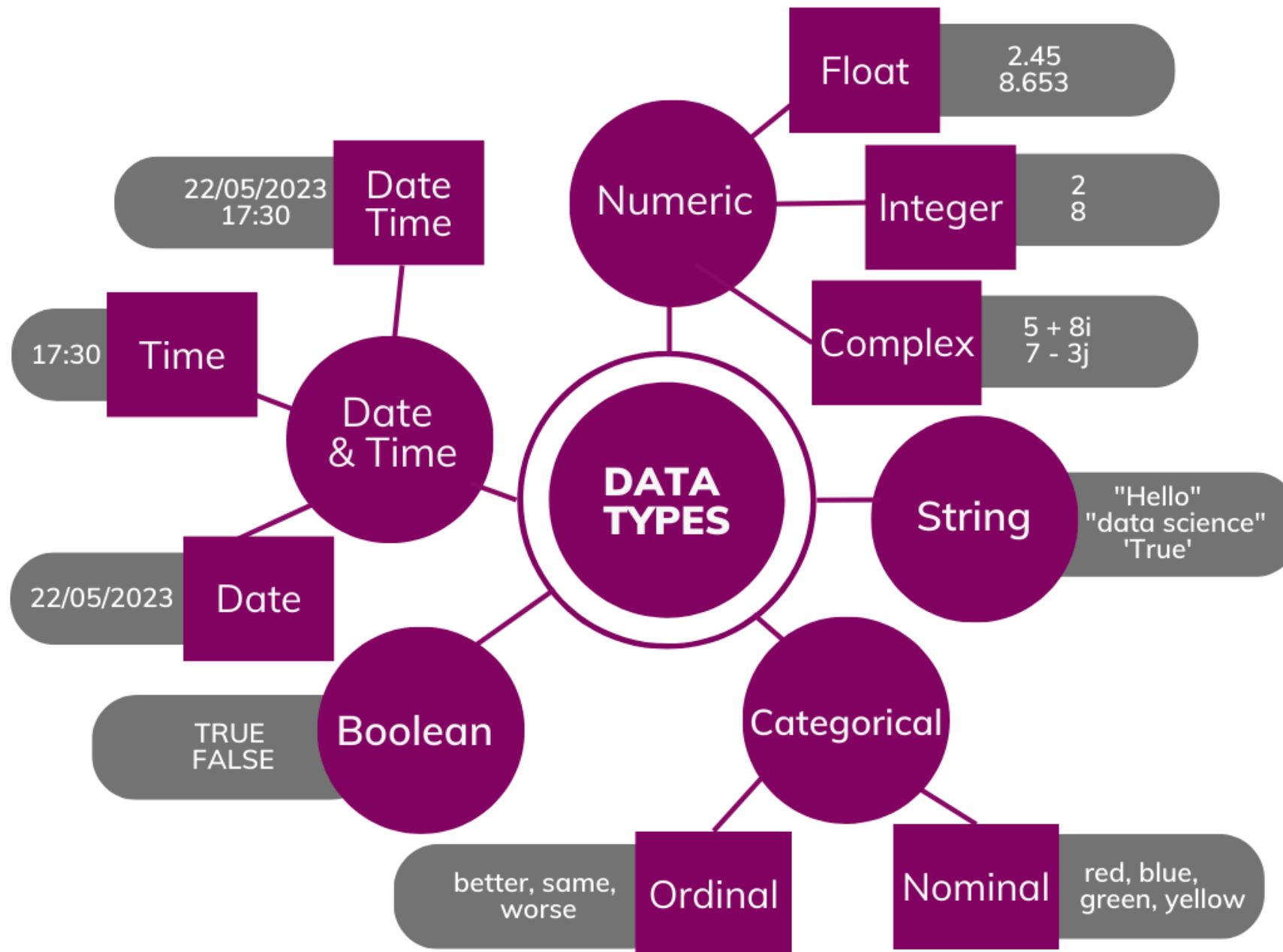
Error in library(gapminder) : there is no package called 'gapminder'

2. stop(packageNotFoundError(package, lib.loc, sys.call()))
1. library(gapminder)

What are data types

- Type of data or values an object contains
 - Internal construct that a programming language uses to understand how to store and manipulate data
- Determines:
 - What kind of mathematical, relational, or logical operation can be applied
 - Which operations can be performed to create, transform, and use the variable in further computation





Data types produced in data generating scenarios

- Emergency service call outs and transfers (e.g., ambulance)
- Smoking behaviours in the community
- Hospital waiting times for non-elective surgeries
- At-home carers in a local council area (local region)
- Any other health and social care situation you can think of!

Data types produced in data generating scenarios

-  20-35 min: Discuss the data generated in your selected scenario
 - What types of data they are
 - What are the possible range of values
 - Could the data type vary depending on your analytic use case? If so, how?
- Share with everyone what you spoke about in your groups
- Be sure to save your document so you can post it on the discussion boards after!

